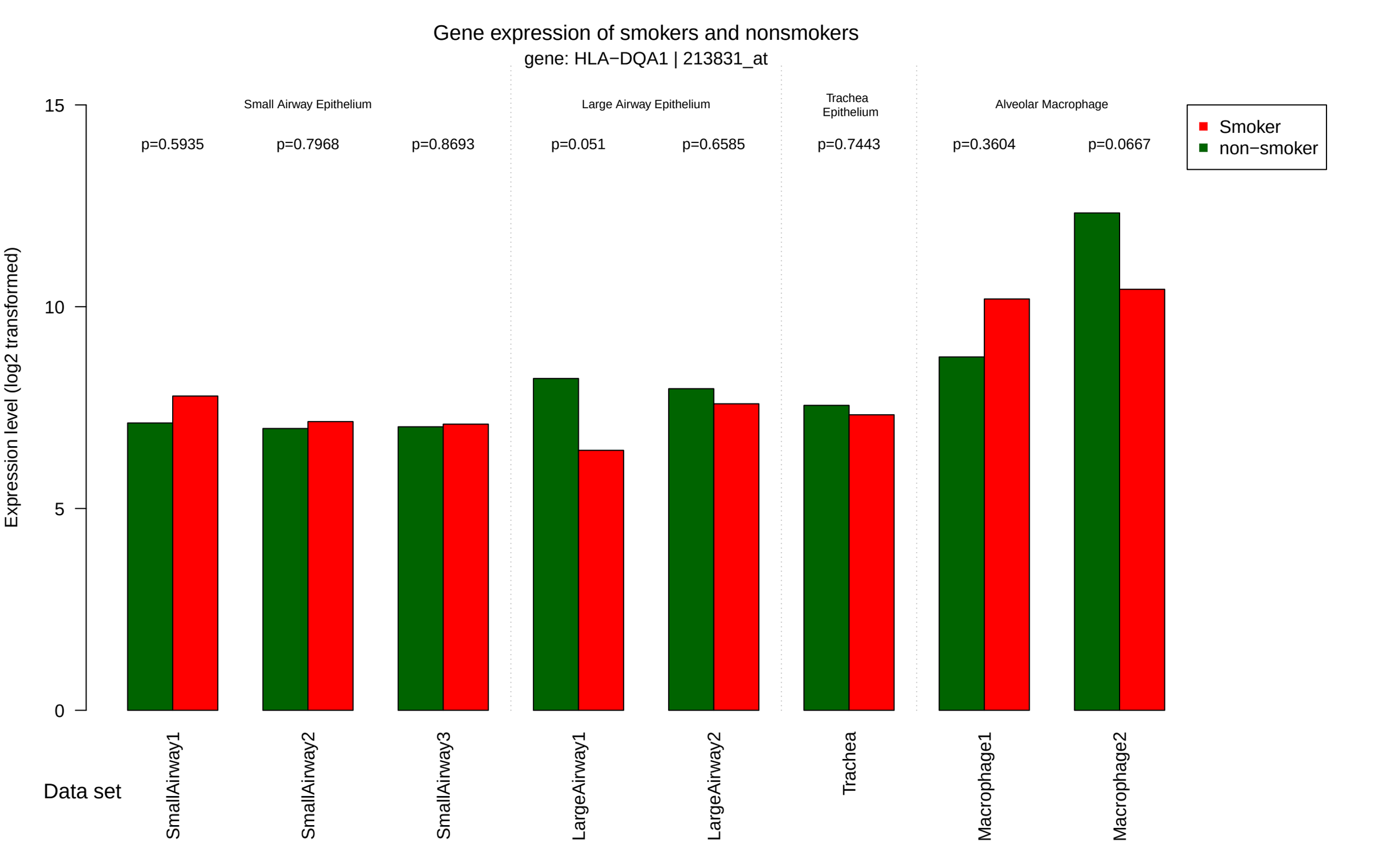
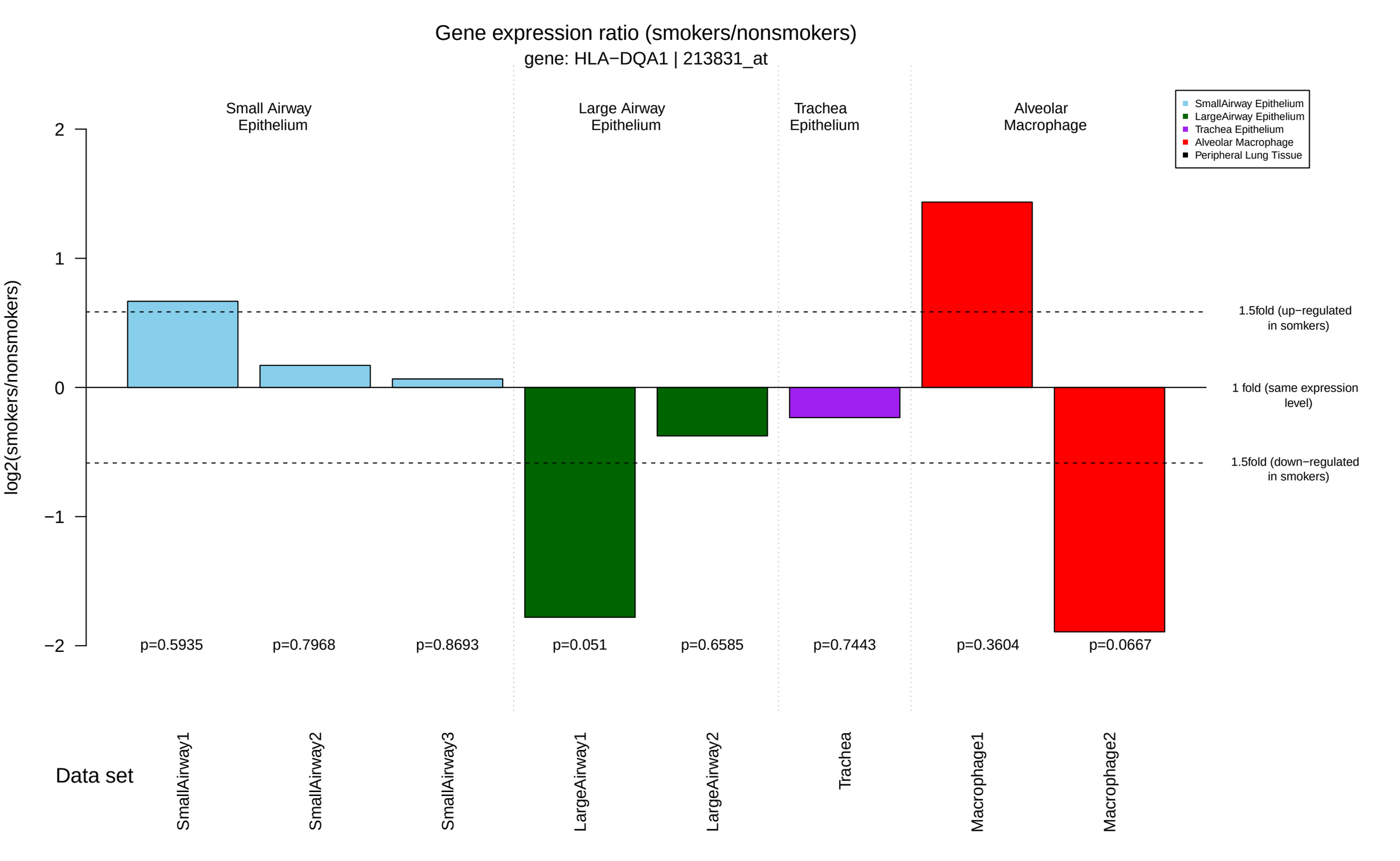
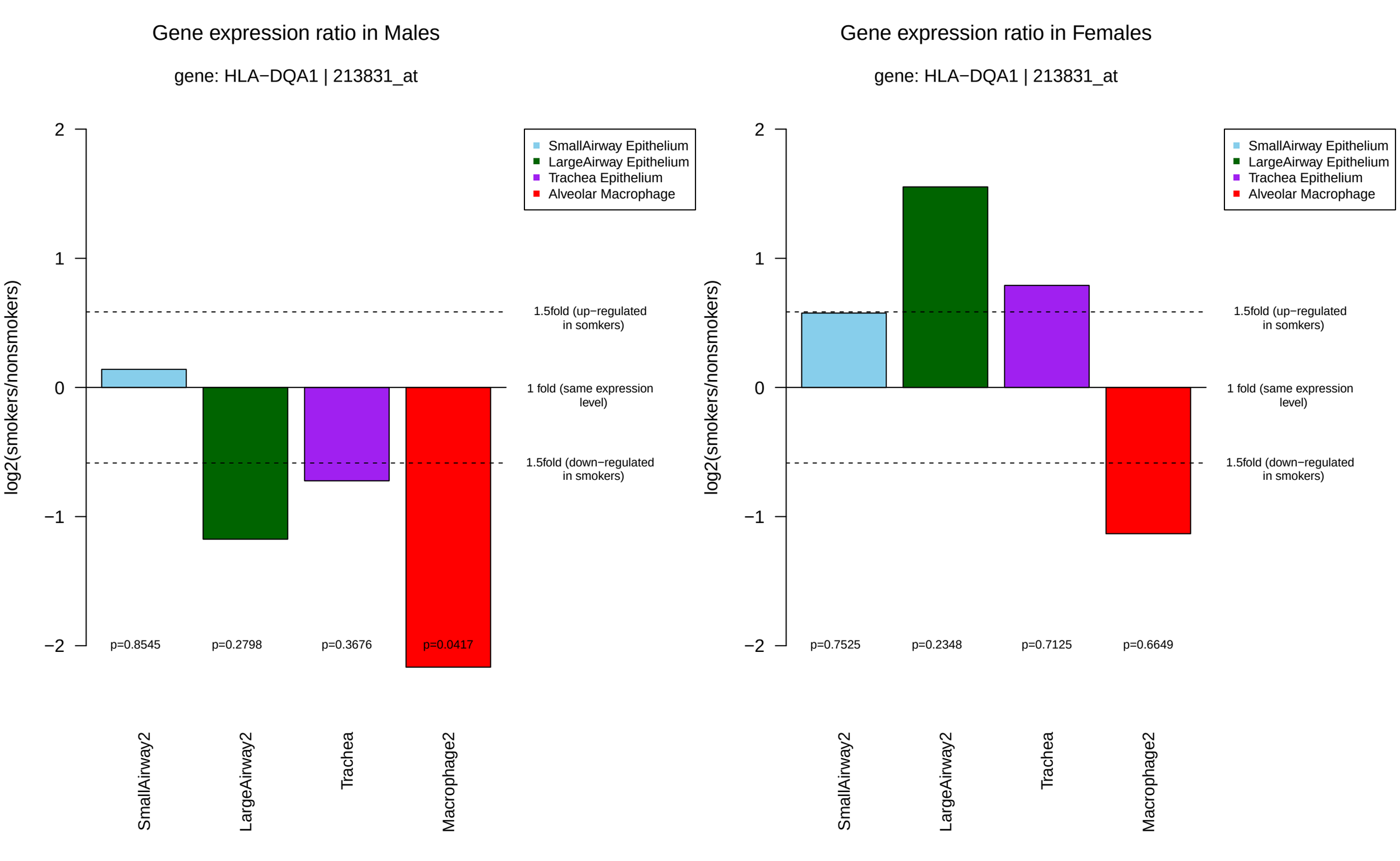
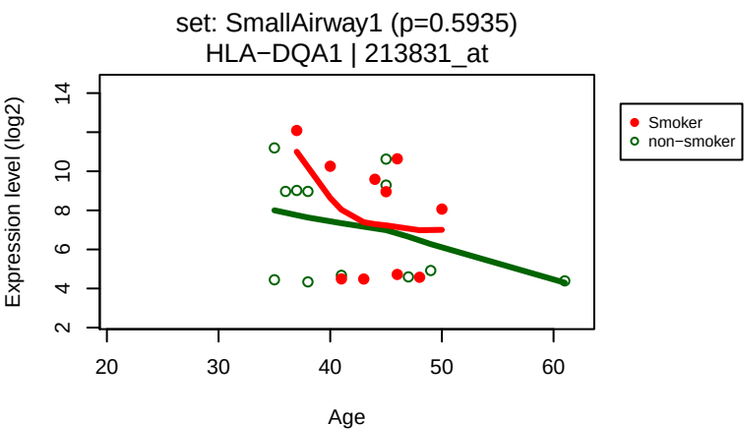
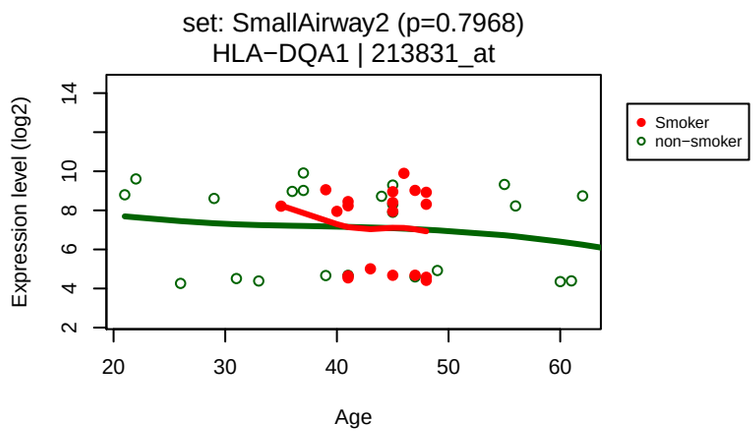
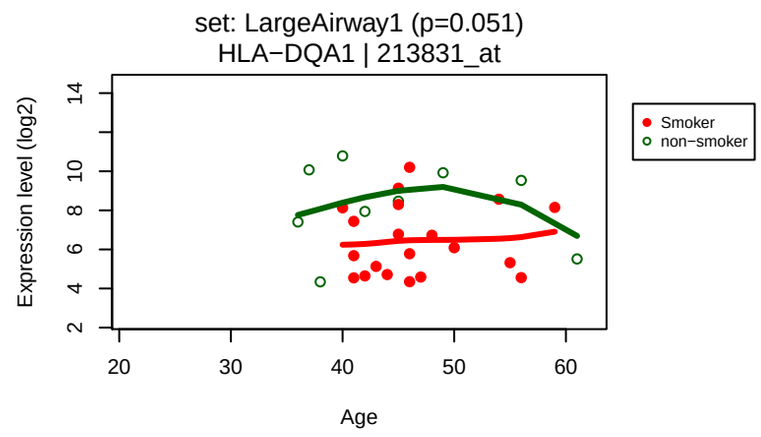

Gene expression of smokers and nonsmokers
gene: HLA−DQA1 | 213831_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5935
p=0.7968
p=0.8693
p=0.051
p=0.6585
p=0.7443
p=0.3604
p=0.0667
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

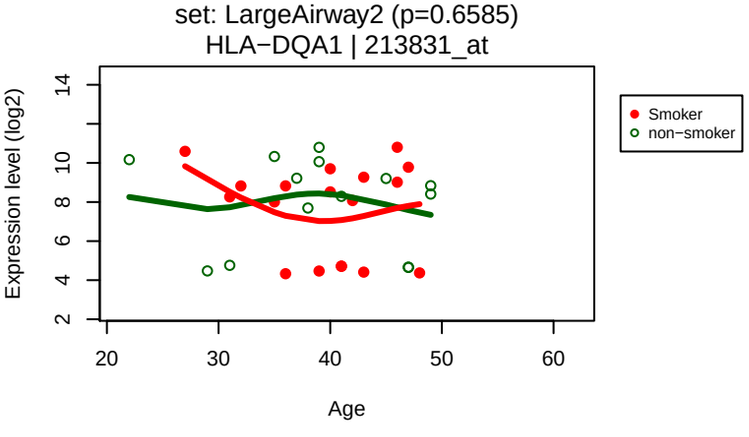

Gene expression ratio (smokers/nonsmokers)
gene: HLA−DQA1 | 213831_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5935
p=0.7968
p=0.8693
p=0.051
p=0.6585
p=0.7443
p=0.3604
p=0.0667
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

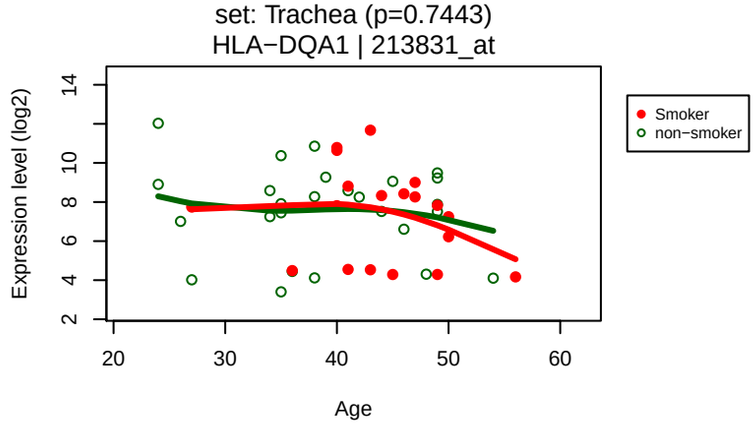

Gene expression ratio in Males
gene: HLA−DQA1 | 213831_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8545
p=0.2798
p=0.3676
p=0.0417
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−DQA1 | 213831_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7525
p=0.2348
p=0.7125
p=0.6649
SmallAirway2
LargeAirway2
Trachea
Macrophage2

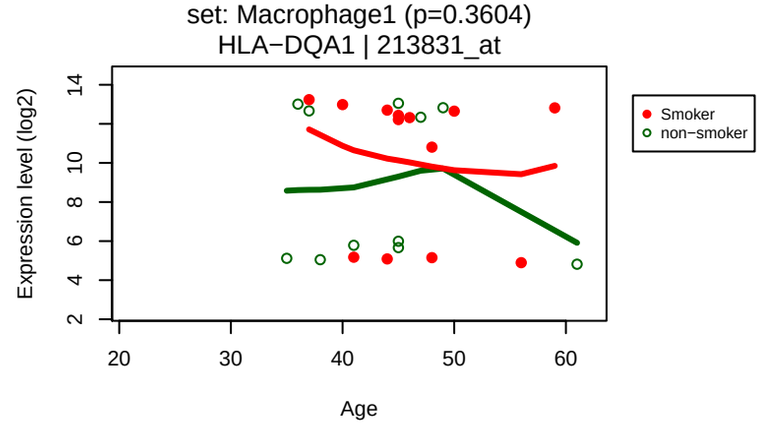

set: SmallAirway1 (p=0.5935)
HLA−DQA1 | 213831_at
Expression level (log2)
Age
Smoker
non−smoker

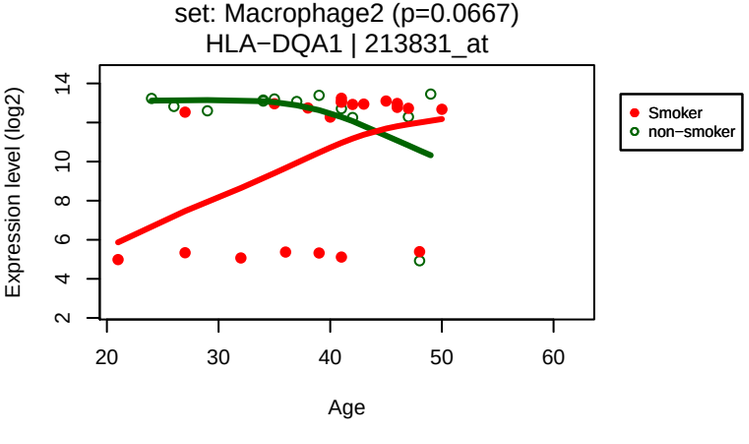

set: SmallAirway2 (p=0.7968)
HLA−DQA1 | 213831_at
Expression level (log2)
Age
Smoker
non−smoker

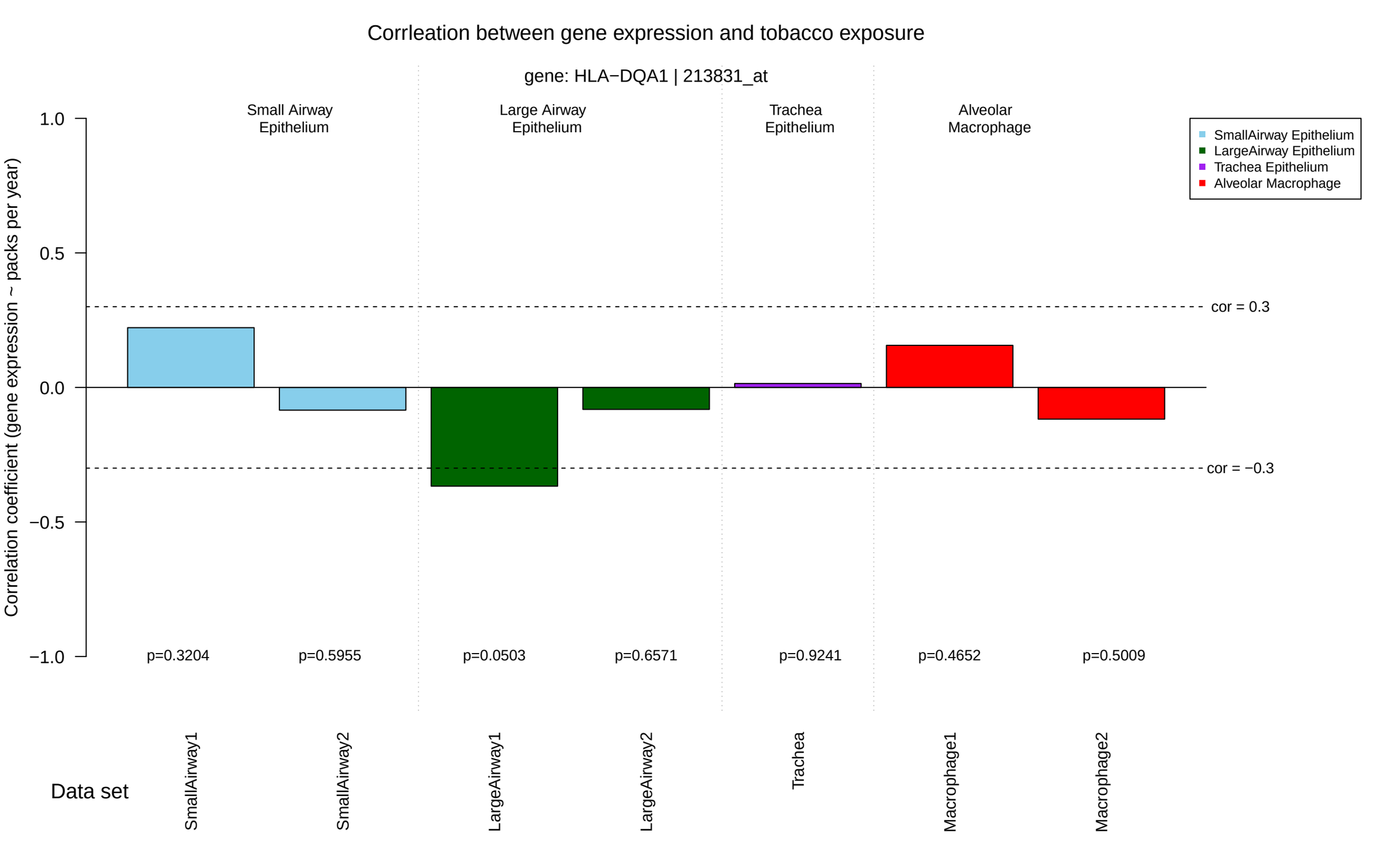

set: LargeAirway1 (p=0.051)
HLA−DQA1 | 213831_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6585)
HLA−DQA1 | 213831_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7443)
HLA−DQA1 | 213831_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3604)
HLA−DQA1 | 213831_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0667)
HLA−DQA1 | 213831_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−DQA1 | 213831_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3204
p=0.5955
p=0.0503
p=0.6571
p=0.9241
p=0.4652
p=0.5009
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2